

Homework 4: MANOVA

S&DS 5360 | Multivariate Statistics

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Setup & Data Overview

We will study the following variables, some of which were manually encoded after data cleaning, and other of which were part of the original study:

- **Continuous Response:** Depression Score (`deprawsc`) and Flourishing Score (`flourish`)
- **Categorical Predictors:** Therapy Yes/No (`ther_any`) and Sexual Orientation (`queer`)
- **Continuous Predictor:** Average Weeknight Hours of Sleep (`sleep_wknight`)

Per feedback from the grader, we omit the code to preprocess the data.

Part 1 | Visualizing Our Categorical Response Variables

We present interaction plots for each of our response variables, `deprawsc` and `flourish`.

```
df_plot <- df %>%
  mutate(
    Therapy_Status = factor(ther_any, levels = c(0, 1),
                           labels = c("No Therapy", "In Therapy")),
    Sexual_Orientation = factor(queer, levels = c(0, 1),
                                labels = c("Heterosexual", "Queer"))
  )

par(mfrow = c(1, 2))

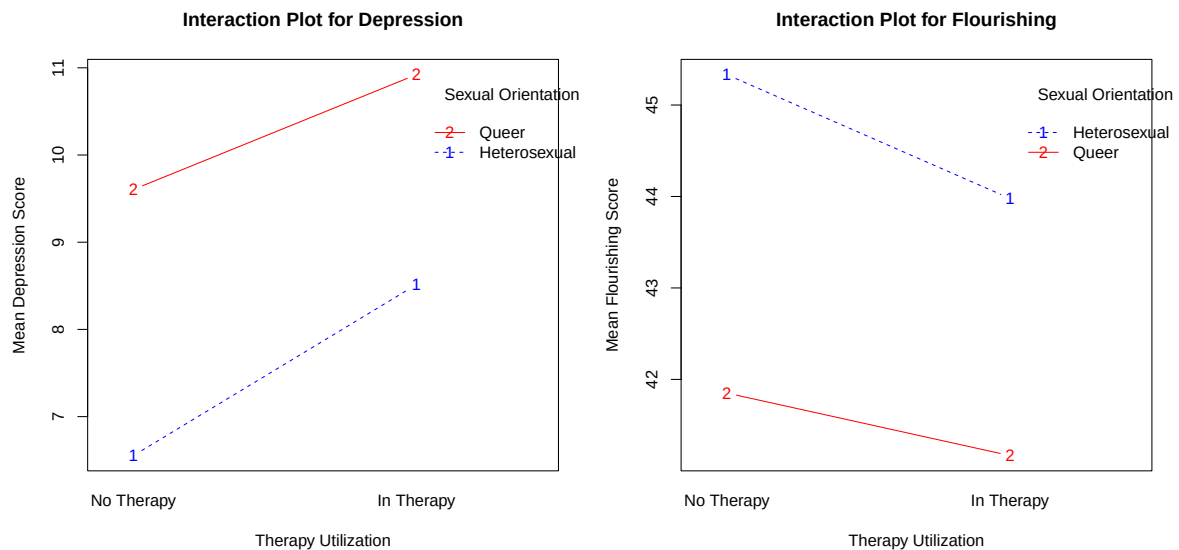
interaction.plot(
  x.factor = df_plot$Therapy_Status,
```

```

    trace.factor = df_plot$Sexual_Orientation,
    trace.label = 'Sexual Orientation',
    response = df$deprawsc,
    fun = mean,
    type = 'b',
    legend = TRUE,
    col = c('blue', 'red'),
    main = 'Interaction Plot for Depression',
    xlab = 'Therapy Utilization',
    ylab = 'Mean Depression Score'
)

interaction.plot(
  x.factor = df_plot$Therapy_Status,
  trace.factor = df_plot$Sexual_Orientation,
  trace.label = 'Sexual Orientation',
  response = df$flourish,
  fun = mean,
  type = 'b',
  legend = TRUE,
  col = c('blue', 'red'),
  main = 'Interaction Plot for Flourishing',
  xlab = 'Therapy Utilization',
  ylab = 'Mean Flourishing Score'
)

```



We observe that across both continuous response variables, we see roughly parallel lines. This indicates there is likely no interaction; that is, the effect of being in therapy or not seems to be equivalent for both queer and heterosexual students.

The positive slope of both lines for depression indicates that therapy utilization is associated with higher levels of depression. By comparison, the negative slope of both lines for flourishing indicates that therapy utilization is associated with lower levels of flourishing, as expected.

Finally, observing the gap between lines in both plots, we determine the main effects. Queer students seem to have lower flourishing scores than their heterosexual counterparts, regardless of therapy utilization. Conversely, heterosexual students yield higher flourishing scores, regardless of therapy utilization.

Part 2 | Two-Way MANOVA

With an understanding of how our variables interact in hand, we proceed to run two-way multivariate analysis of variance (MANOVA) to further examine the effect of therapy and sexual orientation on depression and flourishing. We perform a Type III (Marginal) MANOVA test because it is the most conservative approach for our Likert-scaled survey data with unequal group sizes. It does not assume there is no interaction (Type II), nor does it assume a perfectly balanced experiment (Type I). We test the following hypotheses:

Test 1: Does the effect of therapy depend on sexual orientation?

- H_0 : No multivariate interaction between therapy utilization and sexual orientation; effect on mental health response variables is the same for queer and heterosexual students.
- H_1 : There exists a multivariate interaction; the effect of therapy on mental health response variables differs based on sexual orientation.

Test 2: Do queer and heterosexual students have different mental health profiles?

- H_0 : $\mu_{\text{heterosexual}} = \mu_{\text{queer}}$
- H_1 : $\mu_{\text{heterosexual}} \neq \mu_{\text{queer}}$

Test 3: Do therapy users and non-users have different mental health profiles?

- H_0 : $\mu_{\text{therapy}} = \mu_{\text{no therapy}}$
- H_1 : $\mu_{\text{therapy}} \neq \mu_{\text{no therapy}}$

```
manova_fit <- lm(cbind(deprawsc, flourish) ~ ther_any * queer, data = df)
Manova(manova_fit, type = 'III')
```

Type III MANOVA Tests: Pillai test statistic

	Df	test stat	approx F	num Df	den Df	Pr(>F)
(Intercept)	1	0.95441	317320	2	30318	< 2e-16 ***
ther_any	1	0.01623	250	2	30318	< 2e-16 ***
queer	1	0.03272	513	2	30318	< 2e-16 ***
ther_any:queer	1	0.00065	10	2	30318	5.1e-05 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Using Pillai's Trace, all efforts were statistically significant $p < 0.001$, and we reject all three of the hypotheses noted above. The significance of the interaction (**thera_any:queer**) somewhat contradicts our visual results in Part 1, but we propose that this significance is directly attributable to our large sample size ($n = 30,323$) moreso than the presence of interactions between therapy and sexual orientation. This belief is further supported by review of the F statistics, which reveal that the main effects of sexual orientation ($F_{2,30318} = 513$) and therapy ($F_{2,30318} = 250$) are substantially larger than the interaction ($F_{2,30318} = 10$).

```
summary.aov(manova_fit)
```

Response deprawsc :

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
ther_any	1	36426	36426	1054.651	< 2.2e-16 ***
queer	1	48103	48103	1392.727	< 2.2e-16 ***
ther_any:queer	1	648	648	18.753	1.493e-05 ***
Residuals	30319	1047180	35		

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Response flourish :

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
ther_any	1	21130	21130	276.1359	<2e-16 ***
queer	1	63772	63772	833.3973	<2e-16 ***
ther_any:queer	1	700	700	9.1422	0.0025 **
Residuals	30319	2320040	77		

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

In the univariate setting, we see concurrent results. Sexual orientation maintains a greater effect on depression and flourishing ($F = (1392, 833)$, respectively) than therapy attendance ($F = (1055, 276)$, respectively). The interaction weight is nearly two orders of magnitude lower in both cases.

Collectively, these results suggest an additive model. The lack of a practical interaction indicates that those attending therapy display lower mental health scores (higher depression, lower flourishing), regardless of sexual orientation. Similarly, the disparity in mental health between queer and heterosexual students remains regardless of whether they attend therapy. This implies that while therapy is associated with higher distress, it does not appear to fundamentally alter the gap in mental health observed between queer students and their heterosexual peers.

Part 3 | Contrasts

Now, we perform multivariate and univariate contrasts to compare combinations of factors. We will focus on the main effects, following our findings in Part 2.

```
emm_queer <- emmeans(manova_fit, ~ queer | rep.meas)
emm_ther <- emmeans(manova_fit, ~ ther_any | rep.meas)

test(pairs(emm_queer), joint = TRUE)
```

rep.meas	df1	df2	F.ratio	p.value
deprawsc	1	30319	1347.420	<0.0001
flourish	1	30319	808.121	<0.0001

```
test(pairs(emm_ther), joint = TRUE)
```

rep.meas	df1	df2	F.ratio	p.value
deprawsc	1	30319	485.921	<0.0001
flourish	1	30319	84.757	<0.0001

Multivariate pairwise contrasts revealed statistically significant multivariate differences for both factors, indicating that the overall mental health profile (depression and flourishing scores) differs distinctly from heterosexual students, and likewise for therapy users versus non-users. Because our categorical variables are binary (yes or no), these results are exactly the same as what we observed in Part 2.

```
summary(pairs(emm_queer), adjust = 'none')
```

```
rep.meas = deprawsc:
contrast      estimate      SE    df t.ratio p.value
queer0 - queer1    -2.73 0.0744 30319 -36.707 <0.0001
```

```
rep.meas = flourish:
  contrast      estimate      SE    df t.ratio p.value
  queer0 - queer1      3.15 0.1110 30319  28.427 <0.0001
```

Results are averaged over the levels of: ther_any

```
summary(pairs(emm_ther), adjust = 'none')
```

```
rep.meas = deprawsc:
  contrast      estimate      SE    df t.ratio p.value
  ther_any0 - ther_any1    -1.64 0.0744 30319 -22.044 <0.0001
```

```
rep.meas = flourish:
  contrast      estimate      SE    df t.ratio p.value
  ther_any0 - ther_any1     1.02 0.1110 30319   9.206 <0.0001
```

Results are averaged over the levels of: queer

Univariate contrasts using estimated marginal means revealed that for sexual orientation, queer students reported significantly poorer outcomes: when compared to their heterosexual peers, depression scores were on average 2.73 points worse, and their flourishing scores were 3.15 worse. For therapy utilization, students in therapy reported depression scores 1.64 points worse and flourishing scores 1.02 points worse when compared to those not in therapy. Considering that the depression scale ranges from 0 to 27, these figures represent a change that covers up to 10% of that scale. Similarly, the flourishing scale ranges from 8 to 56. These figures represent a change that covers up to 6.6% of that scale. Statistical significance is observed across all *t*-tests.

Part 4 | Adding a Continuous Variable

We add average hours of sleep per weeknight (`sleep_wknight`) as a continuous predictor. First, we check to ensure there are linear relationships between our covariates and responses.

```
a <- ggplot(df, aes(x = sleep_wknight, y = deprawsc)) +
  geom_point() +
  geom_smooth(method = 'lm') +
  labs(
    title = 'Linearity: Sleep vs. Depression',
    x = 'Hours of Weeknight Sleep',
    y = 'Depression Score'
```

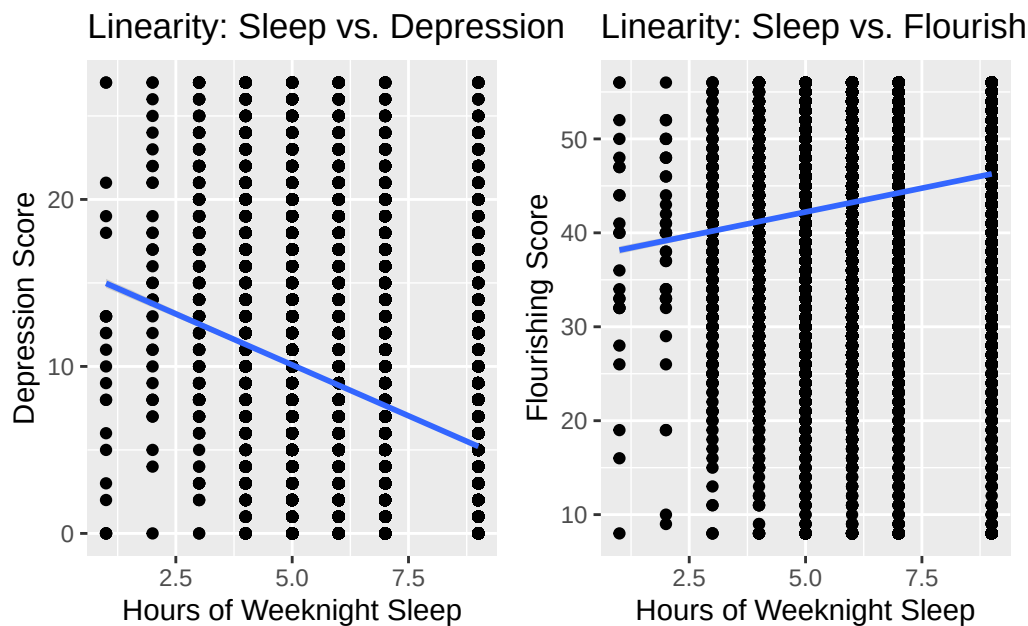
```

)

b <- ggplot(df, aes(x = sleep_wknight, y = flourish)) +
  geom_point() +
  geom_smooth(method = 'lm') +
  labs(
    title = 'Linearity: Sleep vs. Flourish',
    x = 'Hours of Weeknight Sleep',
    y = 'Flourishing Score'
  )

a + b

```



To verify the assumption of linearity, we inspected scatterplots with fitted regression lines. Due to the discrete nature of the sleep variable and the large sample size, the data points appear as vertical bands. However, the fitted regression lines reveal a clear linear trend: as hours of sleep increase, mean depression scores linearly decrease and mean flourishing scores linearly increase. This confirms that modeling sleep as a linear covariate is appropriate for this analysis.

```

model_cov <- lm(cbind(deprawsc, flourish) ~
  ther_any + queer + sleep_wknight, data = df)

```

```
Manova(model_cov, type = "III")
```

Type III MANOVA Tests: Pillai test statistic

	Df	test	stat	approx	F	num	Df	den	Df	Pr(>F)
(Intercept)	1	0.67098	30914.2	2	30318	<	2.2e-16	***		
ther_any	1	0.02124	329.0	2	30318	<	2.2e-16	***		
queer	1	0.04822	768.0	2	30318	<	2.2e-16	***		
sleep_wknight	1	0.08164	1347.5	2	30318	<	2.2e-16	***		

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Somewhat surprisingly, sleep is not only a significant predictor in our multivariate analysis: it is also the largest ($F = 1,348$, opposed to 768 and 329 for queer and therapy, respectively). We can infer that sleep duration explains a unique part of the variance in mental health that is incremental of sexual orientation and therapy use. This does not diminish the robustness of our original predictors; they remain significant with a large magnitude. Said differently, mental health disparities for queer students and students in therapy are not fully explained by sleep differences.

```
summary(model_cov)
```

Response deprawsc :

Call:

```
lm(formula = deprawsc ~ ther_any + queer + sleep_wknight, data = df)
```

Residuals:

	Min	1Q	Median	3Q	Max
	-13.949	-4.162	-1.091	3.261	23.218

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	14.49150	0.15834	91.52	<2e-16 ***
ther_any1	1.73910	0.06793	25.60	<2e-16 ***
queer1	2.64733	0.07098	37.30	<2e-16 ***
sleep_wknight	-1.18990	0.02298	-51.79	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 5.635 on 30319 degrees of freedom
Multiple R-squared: 0.1499, Adjusted R-squared: 0.1498
F-statistic: 1781 on 3 and 30319 DF, p-value: < 2.2e-16

Response flourish :

Call:

```
lm(formula = flourish ~ ther_any + queer + sleep_wknight, data = df)
```

Residuals:

	Min	1Q	Median	3Q	Max
	-39.599	-4.477	1.483	5.569	17.509

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	38.77802	0.24277	159.73	<2e-16 ***
ther_any1	-1.12237	0.10415	-10.78	<2e-16 ***
queer1	-3.08557	0.10883	-28.35	<2e-16 ***
sleep_wknight	0.98013	0.03523	27.82	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

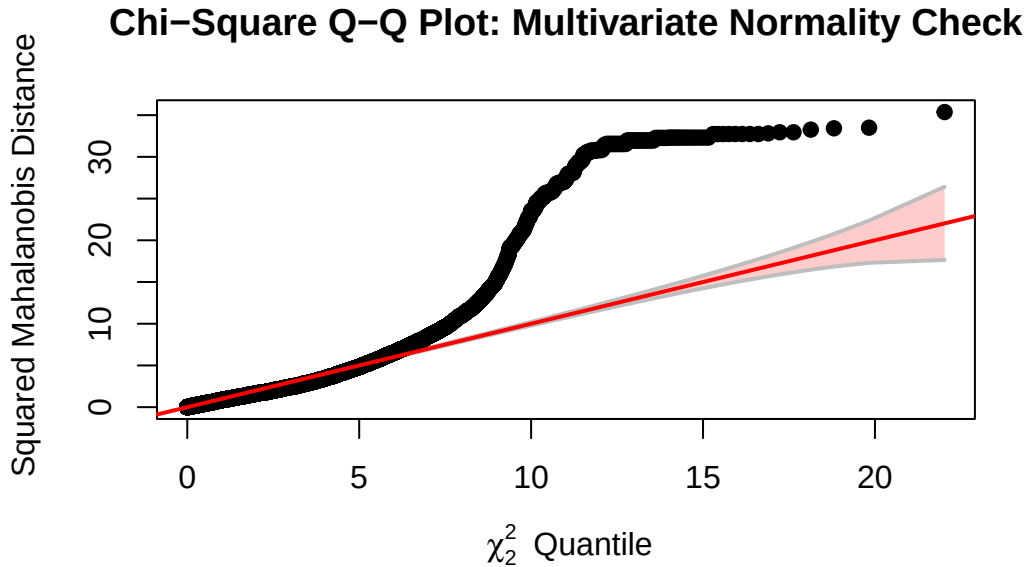
Residual standard error: 8.639 on 30319 degrees of freedom
Multiple R-squared: 0.05931, Adjusted R-squared: 0.05922
F-statistic: 637.2 on 3 and 30319 DF, p-value: < 2.2e-16

Analyzing the univariate results provides the direction and magnitude of the effects in our model. The interpretation of coefficients for sexuality and therapy do not too severely differ from our analysis in Part 2, so we focus on sleep. We see that for every additional hour of sleep a student gets, their depression score drops by 1.19 points, and their flourish score increases by 0.98 points. For context, if a queer student sleeps 2 additional hours a night, this nearly offsets the increase in depression seen amongst queer students.

Part 5 | Evaluating Model Assumptions

To evaluate the assumption of multivariate normality, we produce a Chi-Square Q-Q plot to compare squared Mahalanobis distances of our residuals against the theoretical quantities of a χ^2 distribution with $df = 2$.

```
par(mfrow=c(1,1))
cqplot(model_cov, main = "Chi-Square Q-Q Plot: Multivariate Normality Check")
```



As we have observed in previous analyses of this dataset, our data do not meet the multivariate normality assumption. This is because our data are based on Likert-scale responses, which have a floor effect at 0 and categorical structure. However, as in the past, we have justified this with our significant sample size ($n = 30,320$ for this dataset). By the Central Limit Theorem, the F - and t - tests we ran in the previous sections should remain robust.

However, we consider a square root transformation of our response variables to correct the distribution of our residuals. Previously, we have considered square root and log transformations to correct for the skew, to no avail. For this assignment, we vary our approach by performing a Box-Cox analysis to find the optimal λ for normality.

```
summary(p1 <- powerTransform(cbind(deprawsc, flourish) ~
  ther_any + queer + sleep_wknight, data = df, family="yjPower"))
```

```
yjPower Transformations to Multinormality
      Est Power Rounded Pwr Wald Lwr Bnd Wald Up Bnd
deprawsc    0.4307      0.43    0.4185    0.4430
flourish    2.6711      2.67    2.6265    2.7157
```

```

Likelihood ratio test that all transformation parameters are equal to 0
              LRT df          pval
LR test, lambda = (0 0) 30234.05  2 < 2.22e-16

```

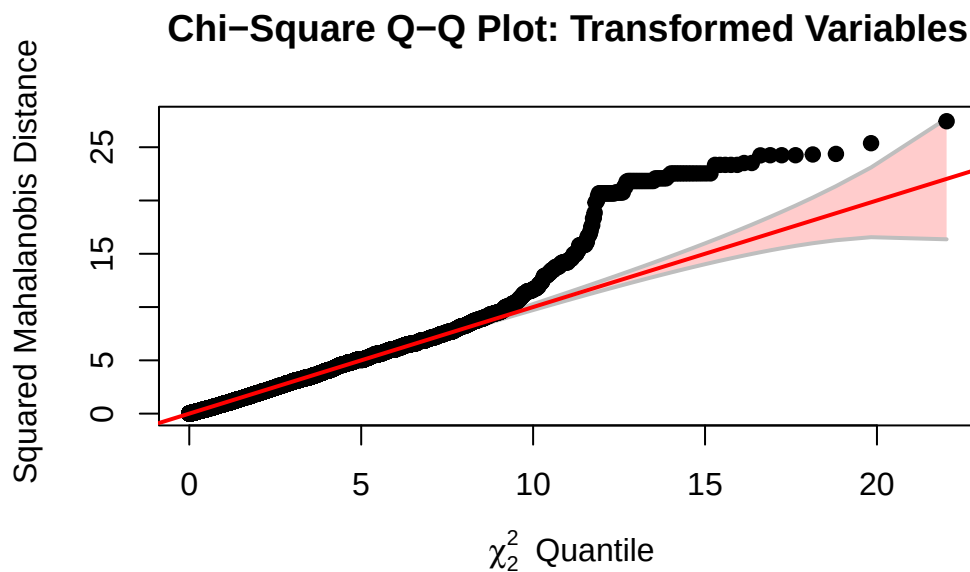
Box-Cox suggests raising depression to 0.4307 and flourishing to 2.671.

```

model_trans <- lm(cbind(df$deprawsc^0.4307, df$flourish^2.671) ~
                  ther_any + queer + sleep_wknight, data = df)

cqqplot(model_trans, main = "Chi-Square Q-Q Plot: Transformed Variables")

```



We see that this does not completely correct for the non-normality, but the heavy tail is more controlled than with our original data. While we do not expect to see any changes in significance for our full model (again, due to our large sample size and extremely small p -values), we re-run the MANOVA on our transformed variables.

```

model_trans <- lm(cbind(df$deprawsc^0.4307, df$flourish^2.671) ~
                  ther_any + queer + sleep_wknight, data = df)

Manova(model_trans, type = "III")

```

```

Type III MANOVA Tests: Pillai test statistic
              Df test stat approx F num Df den Df      Pr(>F)
(Intercept)    1   0.54686  18294.1      2  30318 < 2.2e-16 ***
ther_any       1   0.02530   393.5      2  30318 < 2.2e-16 ***
queer          1   0.05102   815.0      2  30318 < 2.2e-16 ***
sleep_wknight  1   0.08108  1337.6      2  30318 < 2.2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```

As expected, all of our predictors maintain significance, and the relative effect sizes by predictor remain congruent. We see a slight increase in the F -statistic for **queer**, (768 $\rightarrow$ 815) as well as **ther_any** (329 $\rightarrow$ 394), potentially a result of reducing the noise in our data via transformations. On the other hand, **sleep_wknight** remains consistent.

Part 6 | MRPP Test and Adjustment for Multiple Comparisons

Because we conducted multiple univariate tests after our MANOVA, our family-wise error rate increased. We don't expect significance to change after adjusting for multiple comparisons given our extremely lower p -values, but we adjust to exercise proper hygiene. Since our Family-Wise Error Rate has increased, we adjust via all Holm's stepdown method, which is ideal for our larger dataset where we are not sure that our p -values are independent (which would allow us to use Hochberg, the least conservative adjustment method).

```

p_values <- c(2e-16, 2e-16, 2e-16, 2e-16, 2e-16, 2e-16)
names(p_values) <- c("Ther_Dep", "Queer_Dep", "Sleep_Dep",
                    "Ther_Flour", "Queer_Flour", "Sleep_Flour")

p_holm <- p.adjust(p_values, method = "holm"); p_holm

```

Ther_Dep	Queer_Dep	Sleep_Dep	Ther_Flour	Queer_Flour	Sleep_Flour
1.2e-15	1.2e-15	1.2e-15	1.2e-15	1.2e-15	1.2e-15

As expected, given that all p -values were $<2e-16$ before adjustment, we arrive at adjusted p -values that are still tiny: $< 1.2e-15$. This allows us to maintain confidence in the conclusions we drew in Parts 1–5.

Additionally, we perform MRPP, the nonparametric alternative to MANOVA. This is the ideal method, as it does not assume multivariate normality—an assumption our Chi-Square Q-Q plots revealed was violated in Part 5.

```

set.seed(4747)
df_subset <- df[sample(nrow(df), 2000), ]

dist_matrix <- vegdist(cbind(df_subset$deprawsc, df_subset$flourish),
                      method = 'euclidean')

mrpp_queer <- mrpp(dist_matrix, grouping = df_subset$queer)
mrpp_queer

```

Call:

```
mrpp(dat = dist_matrix, grouping = df_subset$queer)
```

Dissimilarity index: euclidean

Weights for groups: n

Class means and counts:

	0	1
delta	11.81	12.81
n	1385	615

Chance corrected within-group agreement A : 0.02714

Based on observed delta 12.12 and expected delta 12.46

Significance of delta: 0.001

Permutation: free

Number of permutations: 999

We take a subset of 2,000 data points, as our full dataset with $n = 30,323$ observations would be computationally intensive to run MRPP on. Our analysis confirms our MANOVA findings, yielding a significant difference ($p = 0.001$) between groups, with chance-corrected within-group agreement of $A = 0.0271$. This confirms the mental health factors (depression and flourishing) of students are significantly more similar within sexual orientation groups than would be expected by chance.

```

mrpp_therapy <- mrpp(dist_matrix, grouping = df_subset$ther_any)
mrpp_therapy

```

Call:

```
mrpp(dat = dist_matrix, grouping = df_subset$ther_any)
```

```
Dissimilarity index: euclidean
```

```
Weights for groups:  n
```

```
Class means and counts:
```

	0	1
delta	12.25	12.47
n	1228	772

```
Chance corrected within-group agreement A: 0.009564
```

```
Based on observed delta 12.34 and expected delta 12.46
```

```
Significance of delta: 0.001
```

```
Permutation: free
```

```
Number of permutations: 999
```

We perform the same analysis on therapy and confirm the same significant difference ($p = 0.001$) between groups, with a chance-corrected within-group agreement of $A = 0.0096$. This smaller A value indicates that while therapy users are distinct from non-users, the distinction between these two groups is pronounced than that of the queer / non-queer disparity identified above.

Collectively, our significant results for both variables confirm our findings in the previous MANOVA sections (Parts 1-6), which maintained their significance in the post-adjustment setting for multiple comparisons. Thus, we conclude that both the usage of therapy and sexual orientation act as distinct and robust determinants of mental health outcomes (depression and flourishing), with disparities in depression and flourishing persisting across parametric and non- parametric evaluations.